

Crack the Code of Serverless Design

Chad Green

Senior Systems Architect

Jasper Engines & Transmissions

Level: Intermediate

Session Survey

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Grew up in software development



Lots of Programming Languages

BASIC

Microsoft BASIC
GW-Basic
QuickBASIC
Visual Basic
VB.NET

C

C
C++
C#
Java

Niche

JavaScript
Pascal
Python
PHP
COBOL

Software development paradigms

Console Applications

Desktop Applications

Distributed Desktop Applications

Web Applications

Distributed Web Applications

COM

DCMO

.NET

Entity Framework

MVC

Service-Oriented Architecture

Microservices

Cloud-Computing

Serverless Computing

Serverless Computing

Functions-as-a-Service

Event-Driven Architecture

Microservices

Work Experience



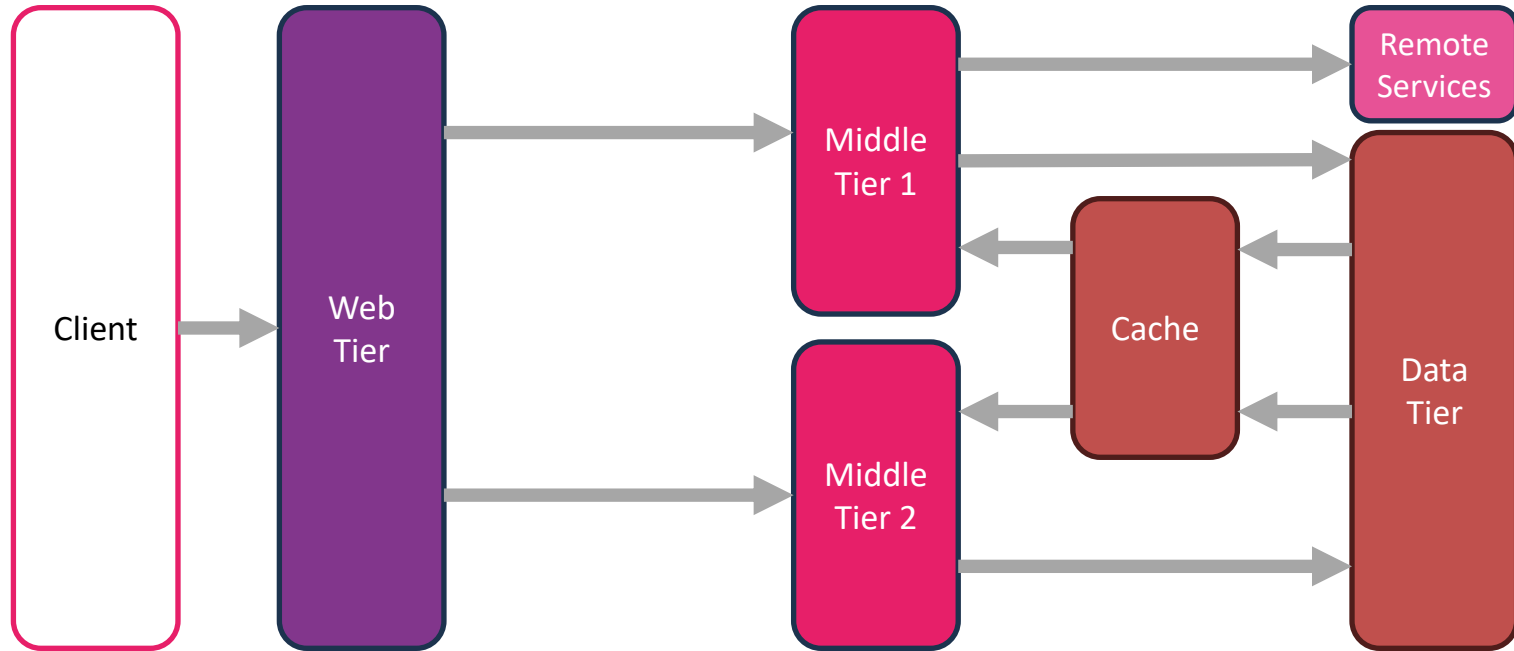
Disclaimer



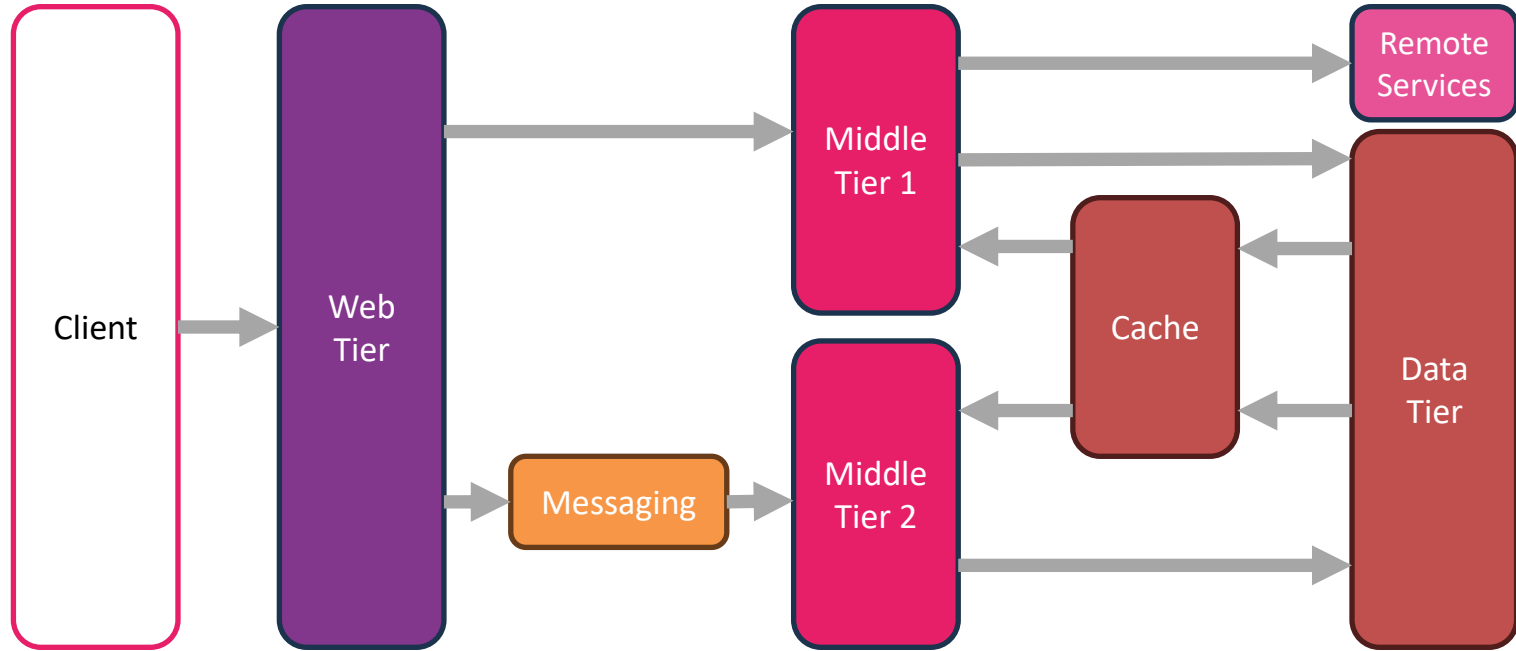
Crack the Code to Scalability

ARCHITECTURE STYLES

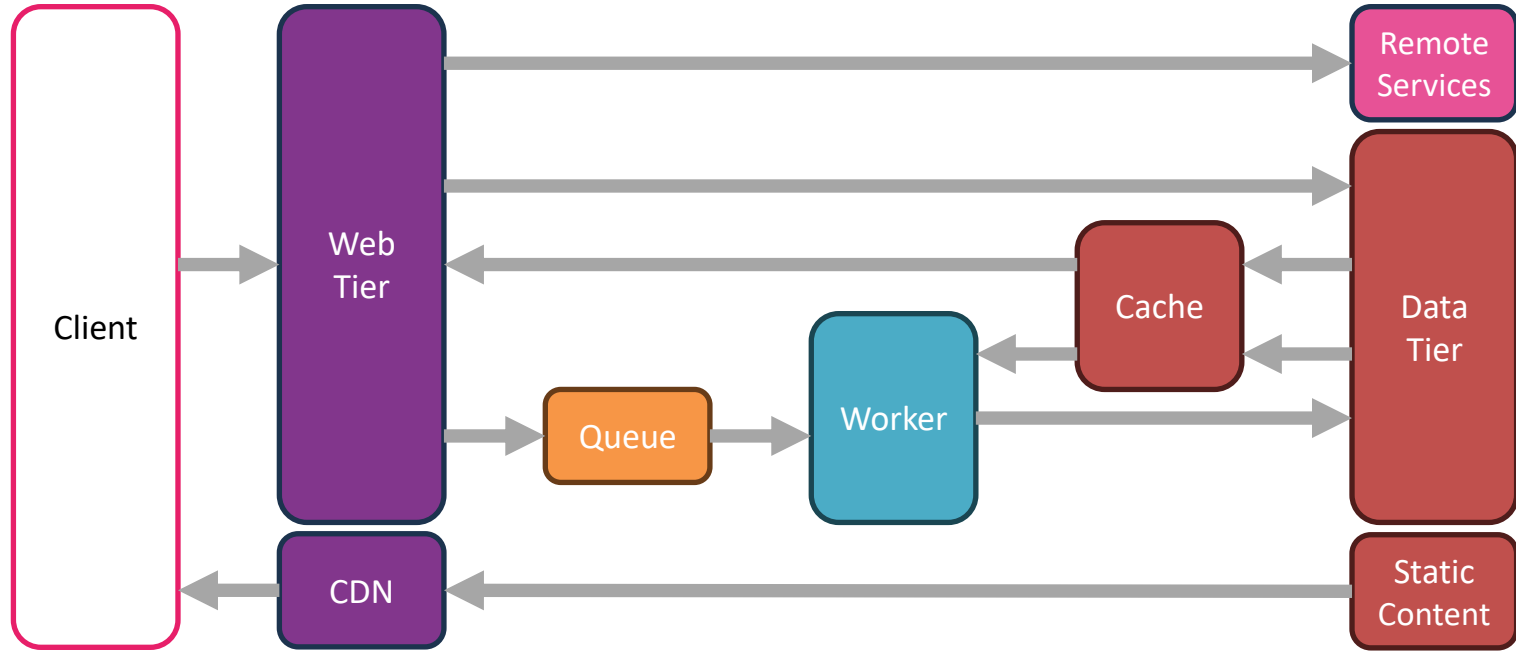
N-Tier



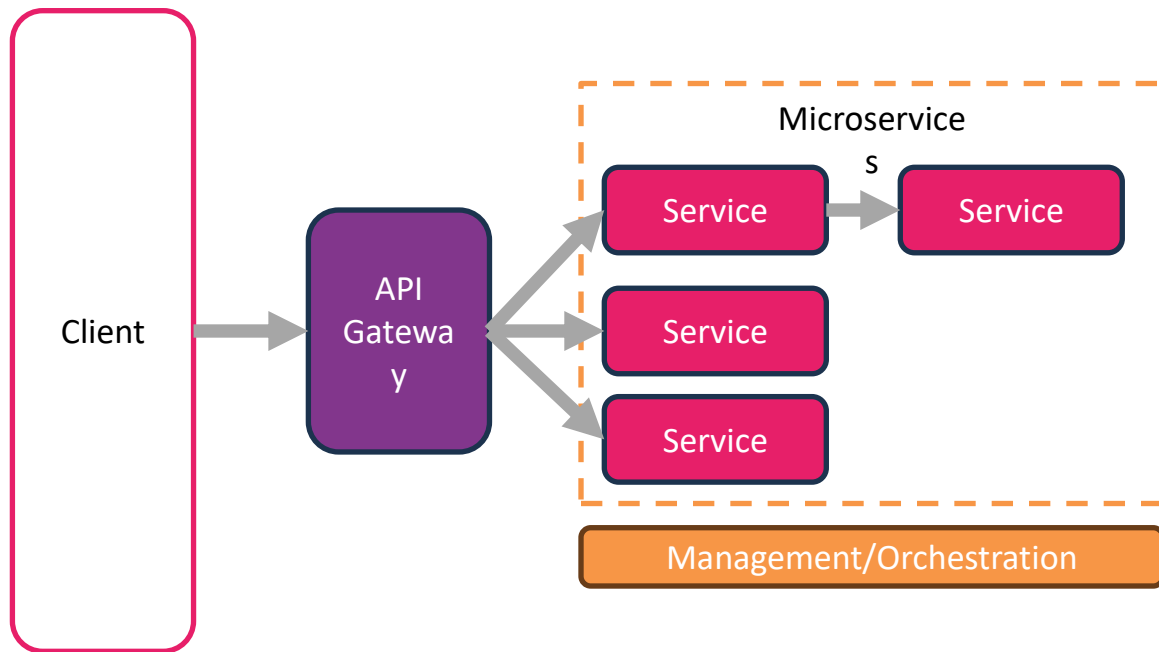
N-Tier with Messaging



Web-Queue-Worker



Microservices



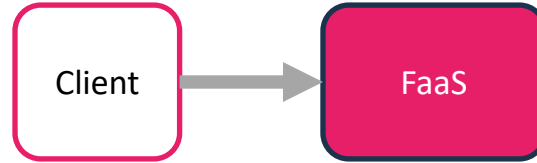
Event-Driven Architecture



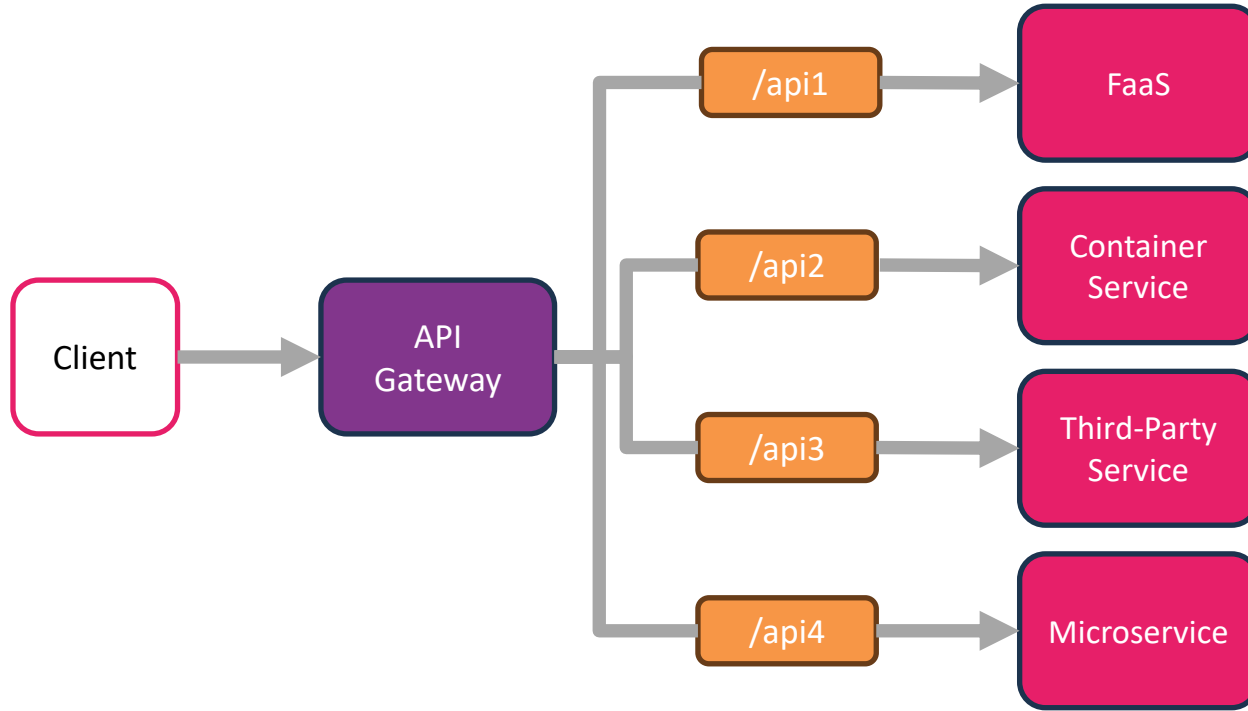
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SERVERLESS DESIGN PATTERNS

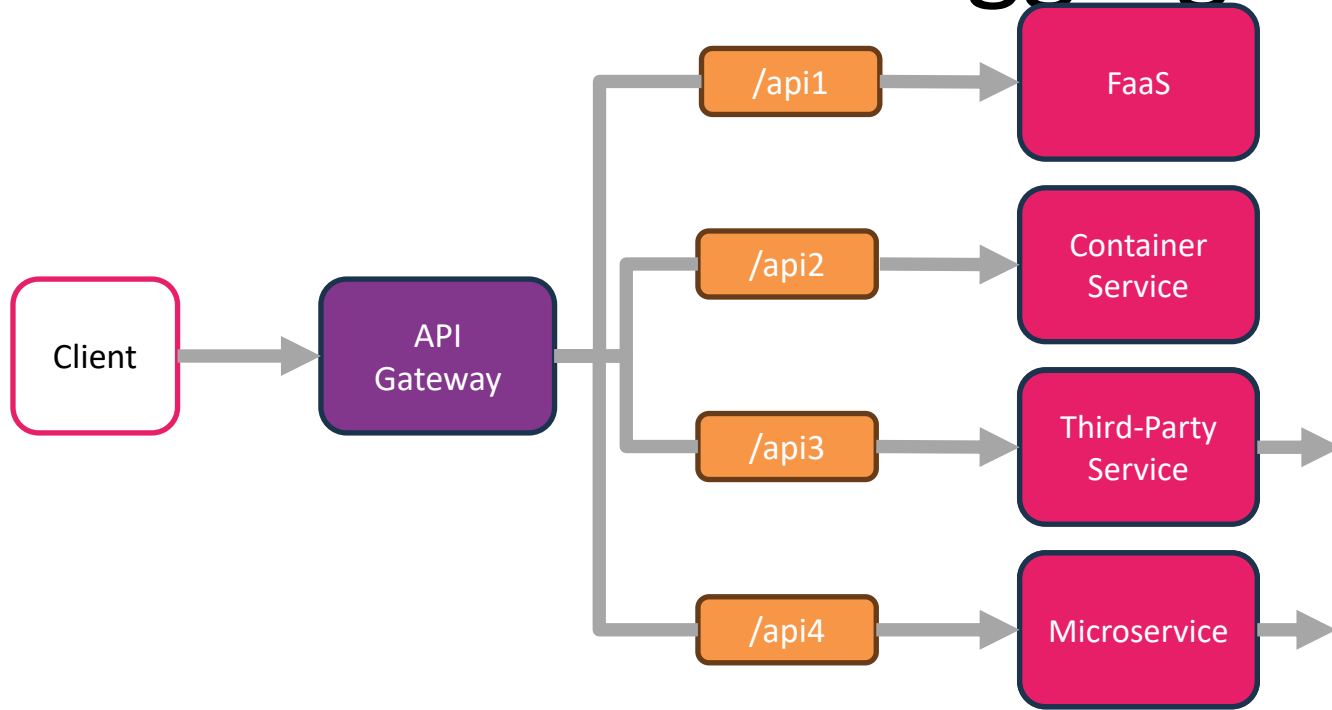
Simple Web Service



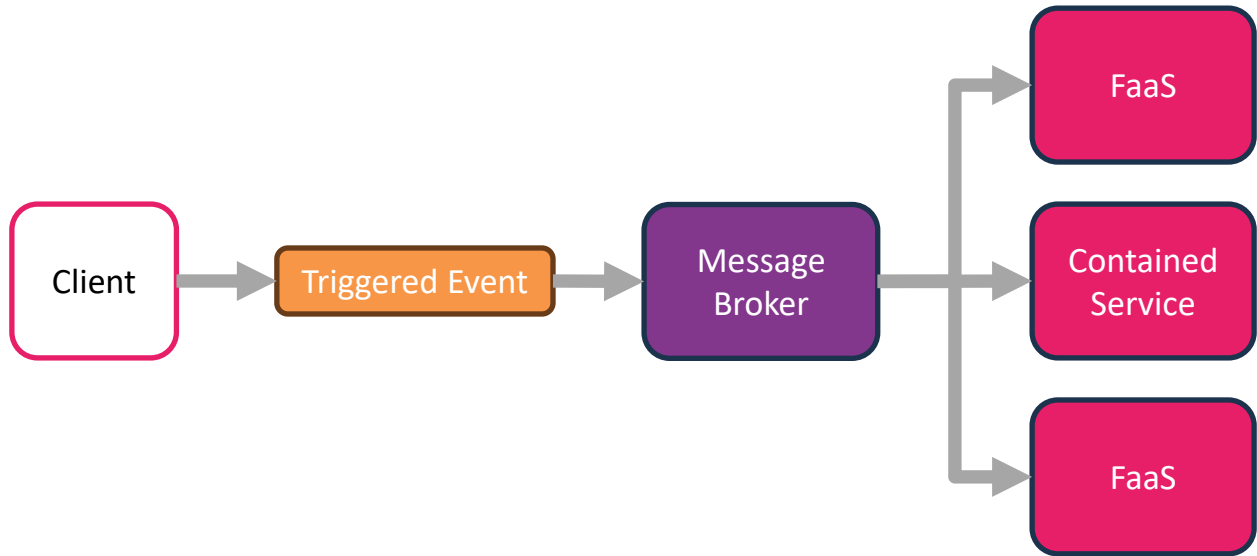
Robust API



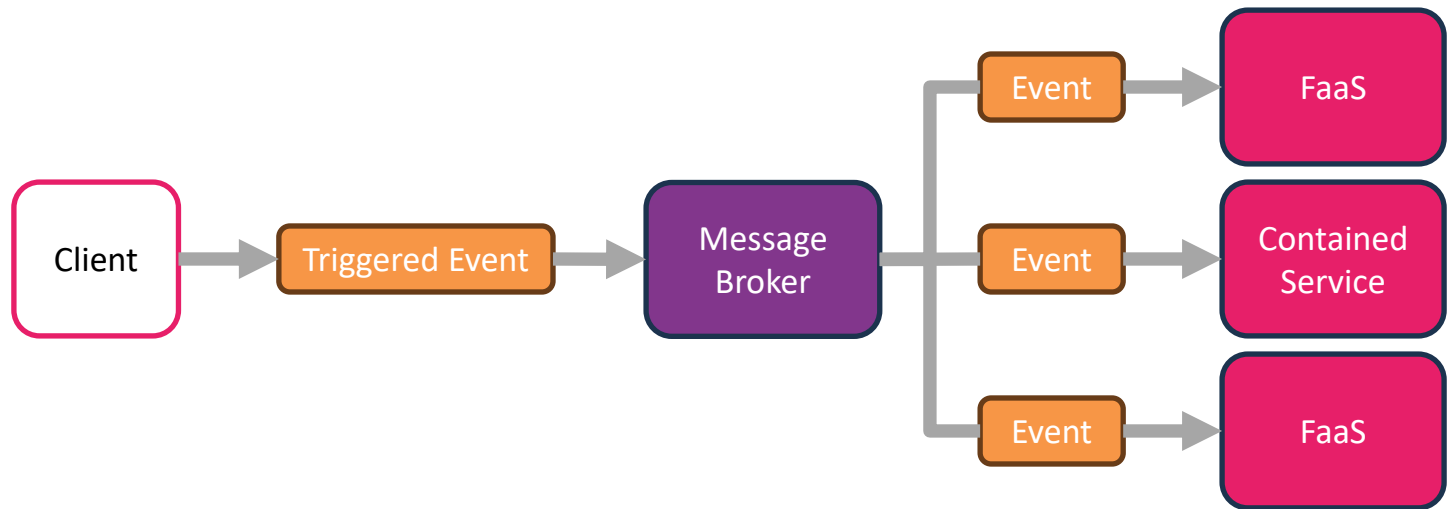
Robust API with Aggregation



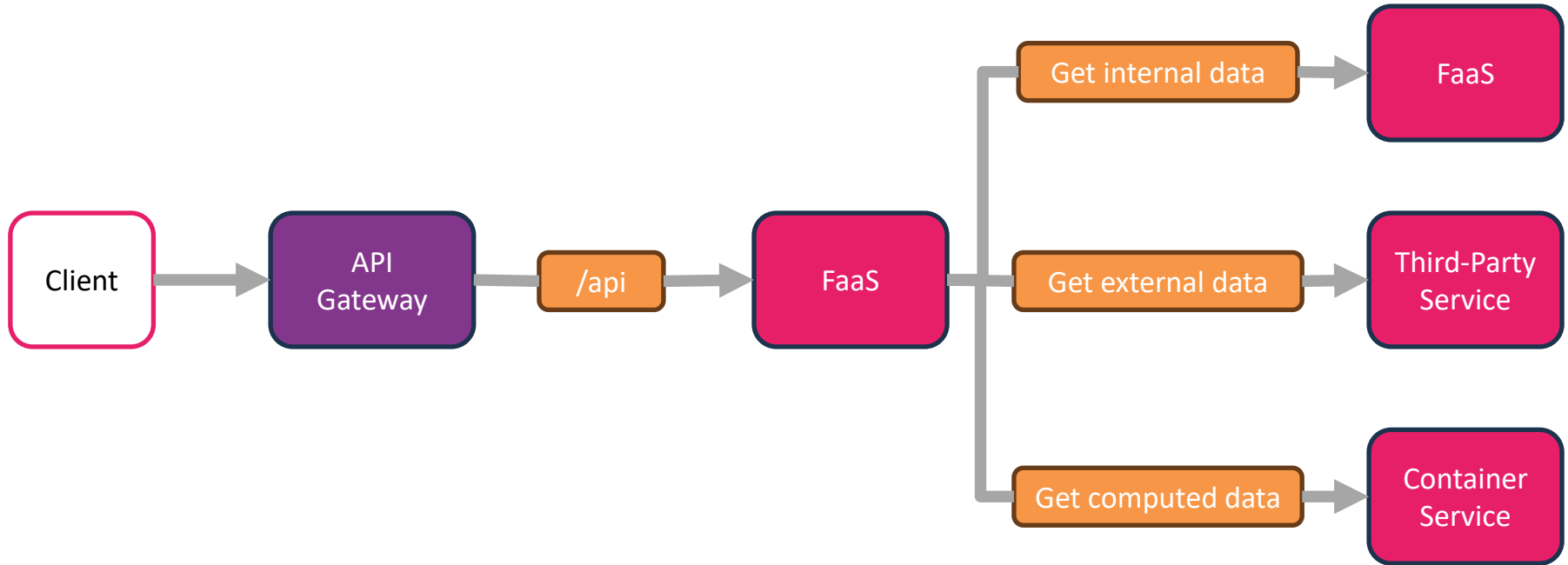
Decoupled Messaging



Publish/Subscribe



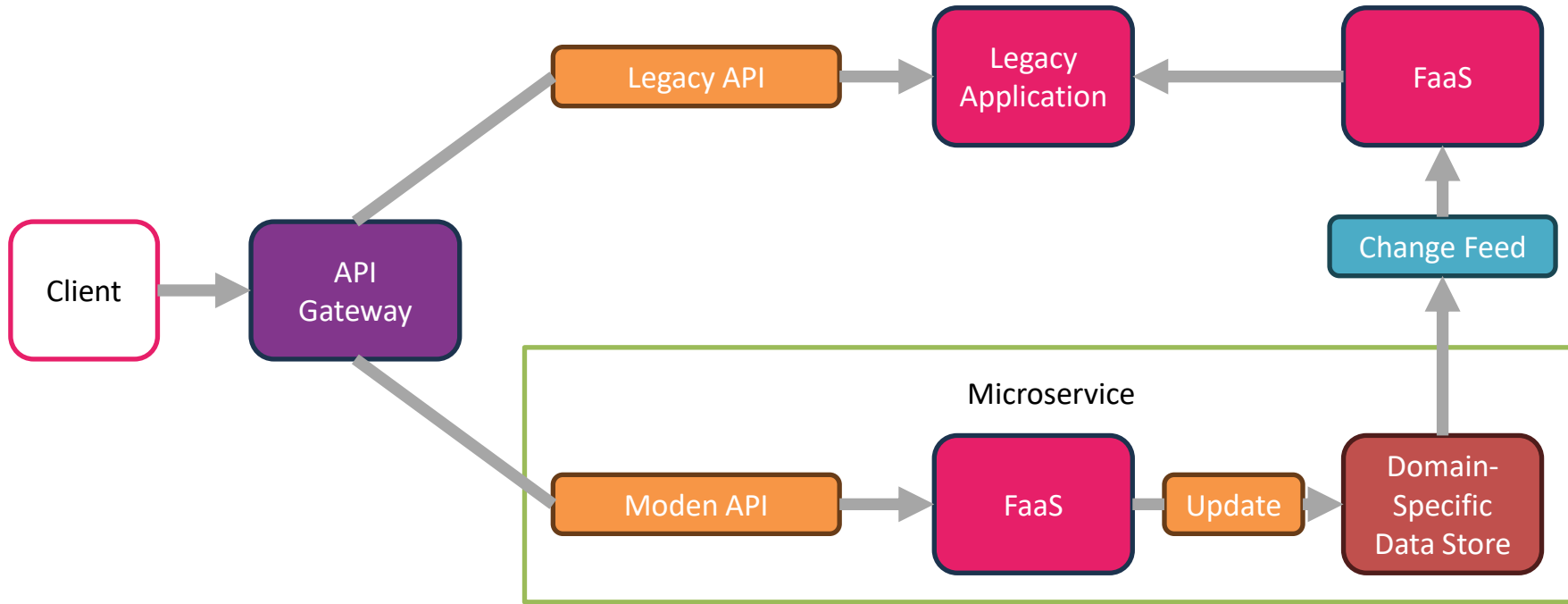
Aggregation



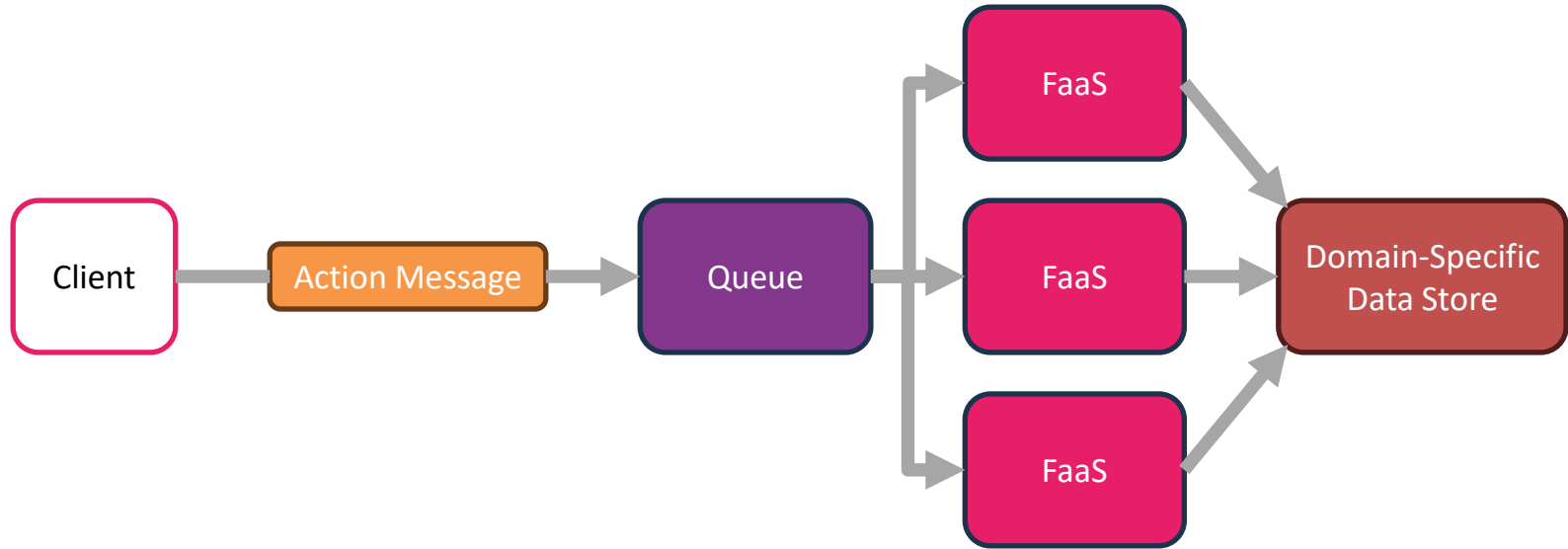
Demo: Aggregation

- A function calling multiple downstream services
- A slow dependency dragging the whole response
- A quick fix (parallel calls, caching, or fallback)

Strangler



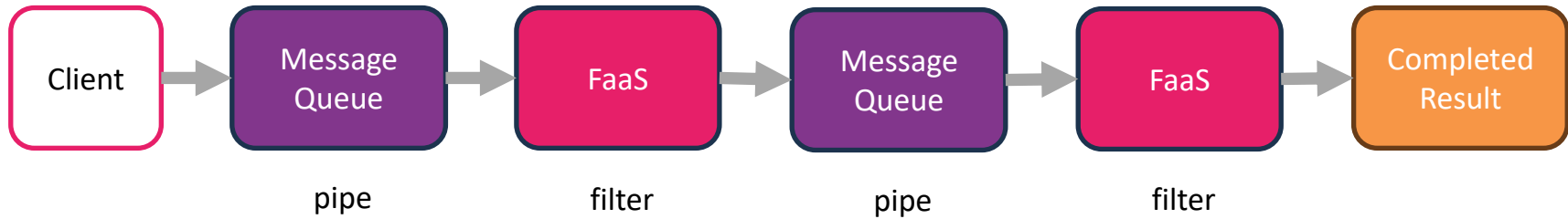
Queue-Based Load Leveling



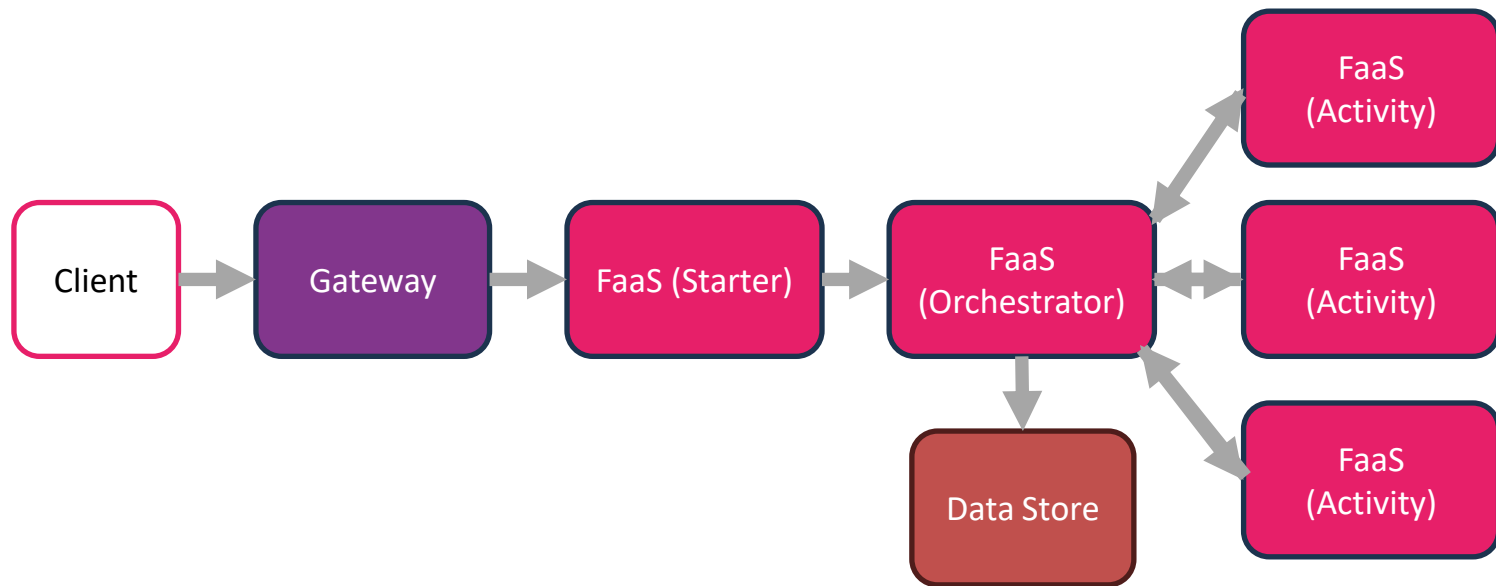
Demo: Queue-Based Load Leveling

- A function that receives a burst of requests
- A queue absorbing the load
- A worker function processing messages at a steady rate
- A poison message scenario

Pipes and Filters



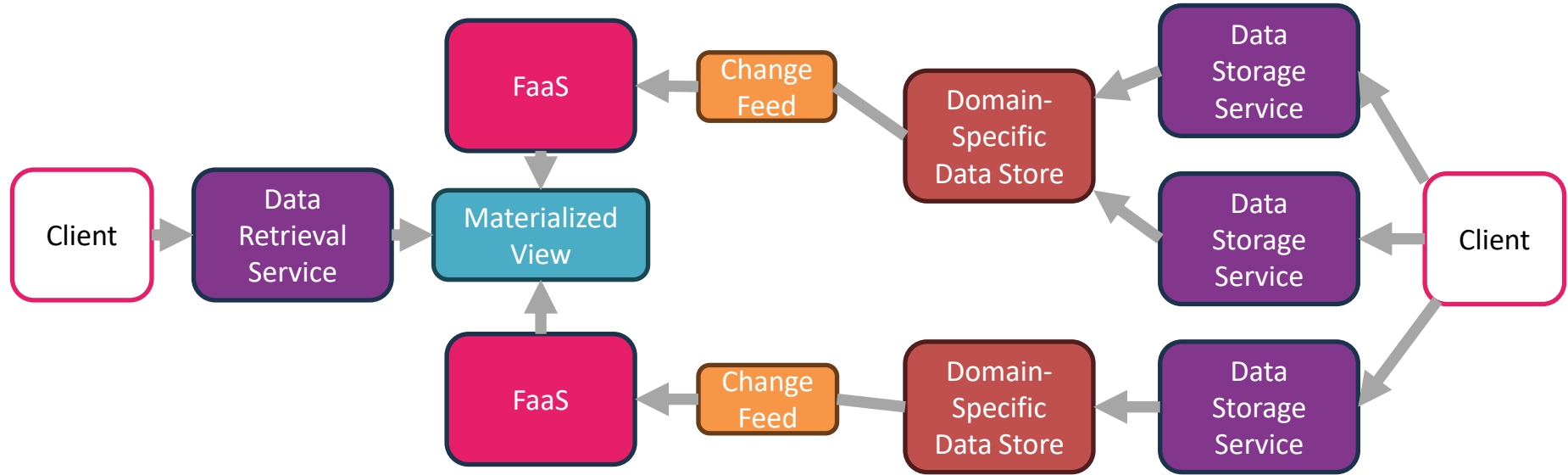
Fan-Out/Fan-In



Demo: Fan-Out/Fan-In

- A starter function kicking off parallel tasks
- A durable orchestrator waiting for all tasks
- A visualization of concurrency
- A quick cost/concurrency discussion

Materialized Views



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PATTERN REALITY CHECK

Pattern Reality Check

Fan-Out/Fan-In

Aggregation

Pub/Sub

Can explode your costs if you are not careful.

Pattern Reality Check

Fan-Out/Fan-In

Aggregation

Pub/Sub

Your API is only as
fast as your slowest
dependency

Pattern Reality Check

Fan-Out/Fan-In

Aggregation

Pub/Sub

Pub/Sub is NOT a
workflow engine.

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SERVERLESS BEST PRACTICES

Serverless Best Practices

Stay Focused

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Handle Errors

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Stay Focused

Handle Errors

Observe What's
Going On

Serverless Best Practices

Stay Focused

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Observe What's
Going On

Test, Test, and
Test Again

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Versioning and
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SHORT INTERACTIVE EXERCISE

You need to process 50,000 incoming images, enrich them with metadata, and store them. Which pattern would you choose?

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SUMMARY AND KEY TAKEAWAYS

Q&A



Stay Connected

✉ chadgreen@chadgreen.com

💬 TaleLearnCode

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